Computer Science

Class 8 (English Medium)

Solved Exercises — Khyber Pakhtunkhwa Textbook Board, Peshawar

Based on Single National Curriculum 2022 — Test Edition

6 units — 80 questions

Unit 1: ICT Fundamentals

Multiple choice questions:

1. The connection of computers in a small geographical area is called:

✓ (a) LAN

(b) MAN

(c) WAN

(d) PAN

Reason: LAN is the interconnection of ICT devices in a small geographical area such as a school, residence or laboratory.

2. In the high rate of distortion in communication environment which of the following communication media is most appropriate?

(a) Coaxial cable

(b) Twisted pair cable

✓ (c) Fiber optic cable

(d) TV cable

Reason: Fiber-optic cable uses pulses of light through glass fiber, giving higher bandwidth and high-speed transmission over longer distances.

3. Which of the following network device transmits message intelligently to the destination device?

✓ (a) Router

(b) Switch

(c) NIC

(d) Access Point

Reason: A router connects two or more LANs and forwards the

incoming message using the best route towards the destination.

4. Which Technology creates computer-generated simulation?

(a) Computer Assisted Translation

(b) Distributed Applications

✓ (c) Virtual Reality

(d) Machine Learning

Reason: Virtual reality creates a computer-generated simulation with objects and scenes that appear to be real.

5. Which technology uses laser beams to create images?

(a) Artificial Intelligence

✓ (b) 3D Holographic Imaging

(c) Robotics

(d) Distributed Applications

Reason: 3D holographic imaging uses laser beams to create three-dimensional images called holograms.

6. Which technology provides telecommunication at high speed?

(a) Virtual Reality

(b) Distributed Applications

(c) Robotics

✓ (d) 5G

Reason: 5G is the 5th Generation cellular network technology developed to increase the speed of telecommunication networks.

Short response questions:

Question: 1. Differentiate between LAN and WAN.

Answer: LAN (Local Area Network) is an interconnection of ICT devices in a small geographical area, such as a school, residence or laboratory. WAN

(Wide Area Network) is a global network of ICT devices that exists over a large-scale geographical area and connects different smaller networks, including LANs; for example, a banking system whose hundreds of branches in different cities are connected to share official data.

Question: 2. What are the benefits of using networks?

Answer: A network is an interconnected system of ICT devices that share hardware and software resources, so users do not need a separate device or program for every computer. In a client-server network the server hosts, delivers and manages most of the resources and services requested by clients, such as print, web and file services, and users can exchange information and communicate with one another through the network.

Question: 3. What is the role of network switch?

Answer: A network switch is a device that allows computers in a network to connect with one another using wired communication. It receives a message from any device connected to the network and transmits that message directly to its destination computer, and it maintains a table (MAC table) for forwarding data. Unlike a hub, which sends data to all computers, the switch sends it only to the destination.

Question: 4. Differentiate between wired and wireless network.

Answer: A wired network uses guided media, that is cables, to transmit data: coaxial cable, twisted pair cable and fiber-optic cable. It gives high-speed, higher-bandwidth transmission over longer distances, but needs physical cabling. A wireless network uses unguided media, that is radio waves, microwave and infrared, with no cables. It is easy to install and maintain, users can communicate while on the move, it provides faster transfer of information and the network is easy to expand.

Question: 5. List different tasks that can be performed by robots.

Answer: Robotics deals with the design, operation and application of robots to perform tasks traditionally done by human beings. According to the book, robots are widely used in automobile manufacturing, in packaging, in laboratory research and in cleaning homes.

Question: 6. Why distributed applications are used?

Answer: A distributed application is computer software installed on servers in a network that runs on multiple computers at the same time to perform a single task. It is used when the workload is too much for a single server to handle. Examples are cellular networks, social networks, online banking and supermarket check-out systems.

Question: 7. What is the purpose of blockchain technology?

Answer: Blockchain is a chain of blocks containing groups of information that cannot be changed or deleted; when information is filled in a block, it is closed and linked to other filled blocks to form a chain. Its purpose is the secure recording and sharing of digital information, and the blockchain database is used in banks and financial services for secure recording of transactions.

Question: 8. Describe 3D holographic imaging.

Answer: 3D holographic imaging is a new technology that uses laser beams to create three-dimensional images called holograms. A 3D hologram displays objects and animations that seem to be real, it exists freely in space, and a person can walk around it and see it from all sides.

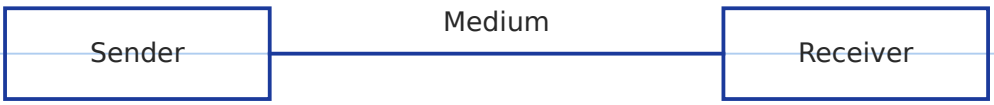
Activities:

Question: 1. Make a physical model of data communication cycle and demonstrate its functionality.

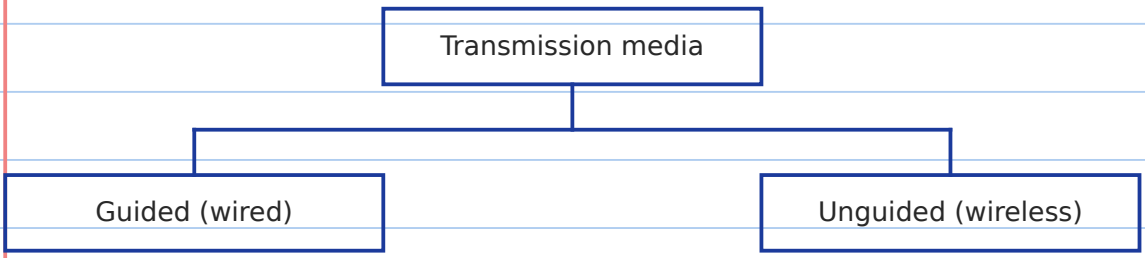
Answer: Build the model with four labelled parts: a sender (for example a computer or phone), a medium (a string or cable joining them), a receiver (a second device), and a message card. Demonstrate it by writing a short message on the card, passing it from the sender along the medium to the receiver, and then having the receiver read it aloud, showing that data communication needs all four parts: sender, message, medium and receiver.

Question: 2. Demonstrate network switch, router, NIC and Wireless Access Point to the students and explain their functions.

Answer: Show each device in turn: the switch connects computers by cable and sends a message directly to its destination computer using its MAC table; the router connects two or more LANs and forwards messages by the best route; the NIC is the card that lets a computer connect to a network to access hardware and software resources including the Internet; and the Wireless Access Point is connected to a wired network and lets devices transfer data wirelessly, extending wireless coverage and increasing the number of users who can connect.



Data Communication System: Sender -> Medium -> Receiver



Guided: Coaxial, Twisted Pair, Fiber-optic | Unguided: Radio waves, Microwave, Infrared

Unit 2: Digital Skills

Multiple choice questions:

1. The basic unit of a worksheet into which you enter data in Excel is called a

✓ (a) Cell

(b) Table

(c) Box

(d) Column

Reason: A worksheet is a collection of cells organized in rows and columns, and data is entered into cells.

2. Which Microsoft Excel function is used to find the highest value in the given range?

Reason: The MAX( ) function is used to find the maximum value from a range of cells, for example =MAX(G4:G8) finds the maximum marks obtained by a student in a class.

3. Formula in Microsoft Excel must start with:

(a) -

✓ (b) =

(c) +

(d) *

Reason: A formula is an expression that operates on values in a cell or a range of cells, and a formula always begins with an equality sign (=).

4. Which Microsoft Excel function checks whether a condition is true or

false?

Reason: The IF( ) function is used to check a condition and returns one value for a TRUE result and another for a FALSE result, for example =IF(I4>=60, "Pass", "Fail").

5. Excel is used to prepare a _____

(a) Presentation

(b) Email

✓ (c) Spreadsheet

(d) Webpage

Reason: Microsoft Excel is a spreadsheet software used to organize and manipulate data using formulas and functions.

6. In a selection, to calculate the number of numeric entries present, _____ function is used.

(a) Sum

(b) Average

✓ (c) Count

(d) Value

Reason: The COUNT function counts the number of numeric entries present in a selection.

Short Response Questions:

Question: 1. Write down the uses of Microsoft Excel.

Answer: Microsoft Excel is a spreadsheet software used for the analysis and storage of data in a tabular form. It is used to organize and manipulate data using formulas and functions, for example preparing a result sheet with obtained marks, averages, percentages and Pass/Fail remarks. It is also used to arrange data in a specific order by sorting, and

to show data as a graph through charts such as column, bar, line and pie charts.

Question: 2. What is the purpose of formula bar?

Answer: The formula bar is the long bar marked with fx above the worksheet. It shows the formula or the contents of the active (selected) cell, as seen in the book's result-sheet figures where the formula bar of cell H4 shows =AVERAGE(C4:F4) and that of cell I4 shows =IF(I4>=60, "Pass", "Fail"). Its purpose is to let the user view and edit the formula or data entered in the selected cell.

Question: 3. Differentiate between worksheet and workbook.

Answer: A workbook is a collection of one or more worksheets in a single file. A worksheet is a collection of cells organized in rows and columns that allows you to organize information as a table. In short, the workbook is the whole Excel file, while a worksheet is one table of cells inside it.

Question: 4. What are the steps to create a Pie chart?

Answer: A chart is a graphical representation of data in a worksheet. The book's steps for creating a chart are: (1) Select the required cells that contain the data to be used in the chart, for example the name and percentage columns of the result sheet. (2) Click the Chart button from the Insert tab. (3) Click the required type of Chart, and for a pie chart choose the Pie chart type. Excel then draws the chart from the selected data.

Question: 5. Label the picture.

Answer: FLAGGED — NOT ANSWERED FROM THE AVAILABLE SCAN. The labelling diagram in the exercise is not legible at scan resolution: the leader arrows point to parts of the Excel window (ribbon, tabs, cells and sheet tabs) but the small print cannot be read reliably, and the book's

Excel-window introduction pages (printed pages 19-20) are missing from the Taleem360 scan. No labels have been guessed. This item must be completed from a complete copy of the book. SCAN-GAP NOTE FOR THIS UNIT: printed pages 19, 20, 23 and 24 are missing from the Taleem360 scan (the printed page sequence jumps from 18 to 21 and from 22 to 25), so Unit 2 content on those pages could not be read; the Unit 2 exercise itself is present and complete apart from this flagged labelling item.

Unit 3: Algorithmic Thinking and Problem Solving

Multiple choice questions.

1. Computers are able to solve problems by performing operations in _____



(a) Nanoseconds

(b) Minutes

(c) Days

(d) Weeks

Reason: Computers solve problems by performing billions of operations per second, so each operation is completed in a tiny fraction of a second.

2. _____ is a graphical representation of an algorithm.

(a) Pie chart

(b) Bar chart



(c) Flowchart

(d) Graph

Reason: One way to provide the flow of the steps of an algorithm is the flowchart, which is a graphical representation of the algorithm.

3. Algorithms are converted to _____ for execution on the computer.

(a) Flowchart

(b) Pseudocode

(c) English language



(d) Program

Reason: After developing an algorithm, the next step is to write the program; a program is intended for the machine, whereas pseudocode

is intended for humans.

4. Repetition is construct that allows the steps of an algorithm to be _____ multiple times.

(a) Tested

✓ (b) Executed

(c) Implemented

(d) Counted

Reason: Loops are used to repeat an action again and again, so the set of code is executed iteratively until the condition is met.

5. Conditional statements are the checkpoints in the programe that determines:

(a) A condition is true

(b) A condition is false

✓ (c) Behavior according to situation

(d) Nature of Algorithm

Reason: Conditions are check points in a program which determine the flow of a program among possible branches based on variable values or events, so the program behaves according to the situation.

6. Value of a variable _____

(a) is fixed

✓ (b) can be changed

(c) depends upon situation

(d) is always based on condition

Reason: A variable refers to a memory location used to hold some data, and that data could be changed during the program execution; a value that does not change is a constant.

7. Nested loops always have _____ number of loops.

✓ (a) two

(b) three

(c) four

(d) as many as required

Reason: A nested loop is a loop inside another loop: an outer loop (Loop1) containing an inner loop (Loop2), as shown in the book's nested-loop pattern.

8. _____ can be directly input into the computers

(a) pseudocode

(b) flowchart

✓ (c) code

(d) algorithm

Reason: Code (the program) is what is entered into the computer for execution; pseudocode is written in plain English for humans, and flowcharts and algorithms are design representations, not machine input.

Short response questions:

Question: 1. Explain the features of information processing model.

Answer: The computer's information processing model works in three stages. First some kind of input is obtained through input devices. Then the input is processed according to some given instructions. Then the output is generated through output devices, and the result may also be stored on storage or network devices. Using this model, computers help us solve problems in a more effective and efficient way, performing billions of operations per second; for larger and complex problems the input, process and output steps are repeated a number of times, producing intermediate

results along the way.

Question: 2. Demonstrate the main steps to solve a problem.

Answer: The book gives six steps to solve a problem: (1) Understand the Problem, (2) Formulate a Model, (3) Develop an Algorithm, (4) Write the Program, (5) Test the Program, and (6) Evaluate the Solution. Before developing the algorithm, the scope of the problem must be defined, and the solution is then written, tested and evaluated against the requirements.

Question: 3. How nested loops are useful in arranging a calendar?

Answer: Nested loops are loops that have loops inside themselves, and they are useful when an iterative task is to be done multiple times. A calendar is arranged by nesting in the same way as the book's clock example: the minute hand goes once around the clock every 60 minutes, making one hour and moving the hour hand. For a calendar, the outer loop runs for each month (or each week) and the inner loop runs for each day of that month, so the inner day loop completes fully for every single step of the outer month loop, filling the whole calendar day by day.

Question: 4. Differentiate between scope and limitations of a problem.

Answer: The scope of a problem defines the boundaries of what a solution must provide to meet the requirements. For example, for a simple calculator the scope is just four functionalities: Addition, Subtraction, Multiplication and Division. The limitation of a problem refers to the aspects that are beyond the study or implementation of the problem; for the same calculator, performing other operations such as percentage, square root, power and log is its limitation. In the email example, a network fault is a limitation (you cannot resolve it at your end), while

checking your own system's network settings is within the scope.

Activities:

Question: 1. Write a pseudocode that tells a user that the number they entered is not a five or a six.

Answer: Line 0: Start; Line 1: Take input (number); Line 2: IF (number = 5 OR number = 6) then: Line 3: Display "The number is a five or a six"; Line 4: ELSE: Line 5: Display "The number you entered is not a five or a six"; Line 6: END IF; Line 7: End.

Question: 2. Write pseudo code that performs the following: Ask a user to enter a number. If the number is between 0 and 10, write the word blue. If the number is between 10 and 20, write the word red. if the number is between 20 and 30, write the word green. If it is any other number, write that it is not a correct color option.

Answer: Line 0: Start; Line 1: Take input (number); Line 2: IF (number >= 0 AND number < 10) then: Line 3: Display "blue"; Line 4: ELSE IF (number >= 10 AND number < 20) then: Line 5: Display "red"; Line 6: ELSE IF (number >= 20 AND number <= 30) then: Line 7: Display "green"; Line 8: ELSE: Line 9: Display "It is not a correct color option"; Line 10: END IF; Line 11: End.

Question: 3. Give pseudocode of rolling a dice program between 2 players, turn by turn. If dice rolls a 6, the same player rolls the dice again.

Answer: Line 0: Start; Line 1: Set player = 1; Line 2: Loop Repeat: Line 3: Display "Player (player), roll the dice"; Line 4: Roll dice; Line 5: Display dice value; Line 6: IF (dice = 6) then: Line 7: Display "Same player rolls the dice again"; Line 8: ELSE: Line 9: IF (player = 1) then Set player = 2 ELSE Set player = 1; Line 10: END IF; Line 11: Until <game over>; Line 12: End.

Question: 4. Develop the pseudocode, that execute the above pseudocode 10 times, count the score of both players and announce the winner or match drawn; in the end.

Answer: Line 0: Start; Line 1: Set score1 = 0; Line 2: Set score2 = 0; Line 3: Set round = 0; Line 4: Loop Repeat: Line 5: Set player = 1; Line 6: Loop Repeat: Line 7: Roll dice; Line 8: Add dice value to the score of the current player; Line 9: IF (dice = 6) then keep the same player ELSE switch player; Line 10: Until both players have had their turn; Line 11: Increment round by 1; Line 12: Until <round = 10>; Line 13: IF (score1 > score2) then Display "Player 1 is the winner" ELSE IF (score2 > score1) then Display "Player 2 is the winner" ELSE Display "The match is drawn"; Line 14: End.

Unit 4: Programming

Multiple choice questions:

1. Which is not a construct of programming languages:

(a) Sequence

(b) Condition

(c) Loop

✓ (d) Business

Reason: The constructs of programming languages are sequence (statements executed in order), condition (if/if-else statements) and loop (repeat/repeat-until/forever); Business is not a programming construct.

2. Loop within a loop is called _____ loop

✓ (a) Nested

(b) Next

(c) Newest

(d) Narrow

Reason: Loops that have loops inside themselves are called nested loops.

3. A loop becomes infinite loop if a condition never becomes _____

✓ (a) TRUE

(b) FALSE

(c) NULL

(d) ZERO

Reason: A repeat until loop requires a matching condition such that as soon as the condition matches the loop terminates; if the condition

never becomes TRUE, the loop never terminates and becomes an infinite loop.

4. Floor( ) function always _____ a number.

(a) Rounds down

(b) Rounds up

(c) Cycles down

(d) Cycles up

Reason: The Floor( ) function always rounds a number down to the nearest whole number below it.

5. Where will be the Sprite located if value of 'var' variable comes out to be 10 in the following code block? (when green flag clicked: set var to pick random 1 to 20; if var < 10 then go to x: 100 y: -100; else go to x: 0 y: 0)

(a) 0,0

(b) 0,100

(c) 100,0

(d) 100,-100

Reason: When var = 10, the condition var < 10 is FALSE, so the 'else' part runs and the sprite goes to x: 0, y: 0. The (100, -100) position is used only when var is less than 10.

Short response questions:

Question: 1. For the following code block a sprite will turn to 30 degrees. How many clicks are required for a sprite to complete a full cycle? (when this sprite clicked: turn clockwise 30 degrees)

Answer: One click turns the sprite 30 degrees. A full cycle (one complete turn) is 360 degrees. Number of clicks required = 360 / 30 = 12 clicks. After 12 clicks the sprite returns to its starting direction.

Question: 2. What is the difference in outputs of the following code-blocks, having 4 backdrops? (Left block: when green flag clicked, go to x: 0 y: 0, forever: repeat 4: switch backdrop to next backdrop, wait 0.5 secs; then move 10 steps. Right block: when green flag clicked, go to x: 0 y: 0, forever: repeat 4: switch backdrop to next backdrop, move 10 steps, wait 0.5 secs.)

Answer: In both blocks the backdrop changes 4 times in each repeat cycle, cycling through all 4 backdrops with a 0.5 second wait. The difference is the movement: in the left block the 'move 10 steps' block is outside the repeat loop, so the sprite changes backdrop 4 times and then moves 10 steps only once per cycle. In the right block the 'move 10 steps' block is inside the repeat loop, so the sprite moves 10 steps with every backdrop change, that is 4 moves (40 steps) per cycle. The right block's sprite therefore travels four times faster and moves while the backdrops change, while the left block's sprite pauses and moves only between backdrop cycles.

Question: 3. For the maze game, duplicate the map 'maze-map' and rename it as 'maze-map2'. Next, change the color of lines in the new map from Blue to Red. Add scripts such that the game becomes a multi-level game with the 2 distinct images.

Answer: Follow these steps in Scratch: (1) In the Backdrops (or Costumes) panel, right-click the 'maze-map' image, choose Duplicate, and rename the copy 'maze-map2'. (2) Open 'maze-map2' in the paint editor, select the blue maze lines and change their color to Red, then save. (3) Add a 'level' variable and set it to 1 at the start. (4) Add scripts so that when the sprite reaches the goal of level 1, the backdrop switches to 'maze-map2', the sprite goes back to the start position, and the level variable is set to 2; the sprite's movement scripts keep working because the walls are now

red. (5) When the sprite reaches the goal on 'maze-map2', show a 'You win' message and stop the game. The two distinct maps (blue lines for level 1, red lines for level 2) make it a multi-level game, as the book notes that Scratch can be used to design multi-level games like a maze game map with corresponding code added and adjusted.

Question: 4. Take a 4-digit number as input for a year and check if the year is a leap year or not. Display your output as: "The year 1979 is not a leap year" or "The year 2020 is a leap year."

Answer: year = int(input("Enter a 4-digit year: ")); if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0): print("The year", year, "is a leap year."); else: print("The year", year, "is not a leap year."); For 1979: 1979 % 4 = 3, so it is not divisible by 4, and the program prints "The year 1979 is not a leap year". For 2020: 2020 % 4 = 0 and 2020 % 100 = 20 (not 0), so the program prints "The year 2020 is a leap year".

Question: 5. Take length and width as input from the user for a quadrilateral and check if it is a square or rectangle.

Answer: length = float(input("Enter length: ")); width = float(input("Enter width: ")); if length == width: print("It is a square."); else: print("It is a rectangle."); A quadrilateral whose length and width are equal is a square; if they are different it is a rectangle.

Question: 6. Write a program in python for the following output (print a triangle): * / * * / * * * / * * * * / * * * * *

Answer: for i in range(1, 6): print("* " * i); This prints row 1 with one star, row 2 with two stars, row 3 with three stars, row 4 with four stars and row 5 with five stars, forming the triangle shown.

Question: 7. Write a program in python which displays square of the

numbers and prints in the form of a triangle, as shown below. (1 / 4 9 / 16 25 36 / 49 64 81 100)

Answer: num = 1; for row in range(1, 5): line = ""; for col in range(row): line = line + str(num * num) + " "; num = num + 1; print(line); The outer loop runs for 4 rows; the inner loop prints 1 square in row 1, 2 squares in row 2, 3 squares in row 3 and 4 squares in row 4, using the numbers 1 to 10 squared in order: 1; 4 9; 16 25 36; 49 64 81 100.

Unit 5: Digital Citizenship

Multiple choice questions:

1. Rules to be followed while using digital technology is known as:

(a) Computer application

(b) Computer interaction

✓ (c) Computer ethics

(d) Computer safety

Reason: Computer ethics means acceptable behavior for using digital technology; it is a set of rules to be followed for moral and social issues while using digital technology, particularly the Internet.

2. The law that protects creative work is:

(a) Internet privacy

✓ (b) Copyright

(c) Plagiarism

(d) Online safety

Reason: Copyright is the branch of law which protects work from unauthorized use by other people; the person who expresses ideas in literary, artistic or any form is the natural owner of that form.

3. Malware that transmits itself automatically over a network to infect other computers is:

(a) Virus

✓ (b) Worm

(c) Adware

(d) Spyware

Reason: A worm is the malware that spreads on its own over a

network from one computer to others, without needing a host file or user action.

4. Malware that pops up advertisement is:

(a) Virus

(b) Worm

(c) Spyware

✓ (d) Adware

Reason: Adware is the malware that displays unwanted pop-up advertisements on the user's computer.

5. To report a cybercrime, you can contact:

(a) www.ndma.gov.pk

✓ (b) www.nr3c.gov.pk

(c) www.fbise.edu.pk

(d) www.nadra.gov.pk

Reason: Cybercrime in Pakistan is reported to the NR3C, the National Response Centre for Cybercrime of the Federal Investigation Agency (www.nr3c.gov.pk; helpdesk@nr3c.gov.pk for queries about a complaint).

6. The volunteers who help in prevention of cybercrime are called:

(a) Cyber friend

(b) Eagle Scout

✓ (c) Cyber Scout

(d) Cyber bully

Reason: Cyber Scouts are devoted volunteers enrolled by Cyber Security of Pakistan from different schools and colleges to combat cybercrime at the grass root level.

Short response questions:

Question: 1. Who is cyber scout? How can he/she assist in prevention of cyber crime?

Answer: A Cyber Scout is a person with the unique aim to promote awareness about cybercrime in society. He or she is trained to identify cybercrime activities and is equipped with adequate preventive knowledge to help fight this menace, understands the risks of virtual socializing, and works to promote cyber safety in his or her surroundings, creating cyber awareness among fellow students, teachers and parents. Cyber Scouts assist in prevention through cybercrime awareness trainings, cybercrime preventive measures, sharing information with NR3C (National Response Centre for Cybercrime), promoting careful and responsible use of cyberspace, directing and helping individuals to report cybercrime, patrolling cyberspace, and encouraging good moral use of technology.

Question: 2. Why we need cybercrime laws?

Answer: Cybercrime refers to any crime committed by means of computer and Internet technology by having unlawful access to others' computers. We need cybercrime laws so that such unlawful access, theft and misuse can be punished, victims have a proper place to report (the NR3C), and users, their privacy, their data and their money are protected by law when they use digital technology.

Question: 3. Why computer systems should be protected from unauthorized access?

Answer: Computer systems should be protected because unauthorized access lets criminals steal or change information and get credit card or bank details for financial benefits, which is what hackers do. Malware also attacks privacy, spies on users, destroys data and causes financial loss.

Protection through passwords, careful use and antivirus software keeps private information, data and money safe.

Question: 4. How creator of copyright work financially benefits from it?

Answer: Copyright makes the creator the natural owner of the form in which the ideas are expressed, and it is the branch of law which protects the work from unauthorized use by other people. Because no one else may legally copy, sell or use the work without the owner's consent, the creator alone can earn from it, for example by selling it or by allowing its use on payment, and others cannot take that income away by copying it freely.

Question: 5. Why digital piracy is a crime?

Answer: Digital piracy means illegally copying digital content in any form without the consent of the owner. It is a crime because it breaks the copyright law that protects a creator's work from unauthorized use: the owner is deprived of the earnings from the work, while the pirate gains from something that is not theirs.

Question: 6. Why hackers break into computer systems?

Answer: A hacker is a person who illegally breaks into others' computer systems for stealing or changing information, or to get credit card or bank details for financial benefits.

Question: 7. How spyware can be harmful?

Answer: Spyware is a malware, that is malicious software, created to attack privacy and for spying, destruction and financial benefits. It secretly watches what the user does on the computer, records private information such as passwords and bank details, and passes it to criminals, which can lead to loss of privacy, theft of identity and loss of money.

Activity:

Question: 1. Install an antivirus software on a computer to demonstrate computer protection from malware.

Answer: (1) Choose a reputable antivirus program and install it from its official website or a trusted source. (2) Update it so it has the latest virus definitions. (3) Run a full system scan and let it quarantine or remove any malware it finds, such as viruses, worms, spyware or adware. (4) Turn on real-time protection and schedule regular automatic scans, and show the class the scan report and the quarantine list. This demonstrates how antivirus software protects a computer from malware, which the book describes as malicious software that threatens privacy, spying, destruction and financial benefits.

Unit 6: Entrepreneurship in Digital Age

Multiple choice Questions:

1. What is the component of business plan?

(a) Marketing Strategy

(b) Sales Strategy

✓ (c) Operations

(d) Digital marketing

Reason: There are five components of a business plan: customer needs, product design, costing, operations and marketing. Operations is one of the listed components.

2. Which of the following is the example of Quality Assurance?

(a) Validation

✓ (b) Verification

(c) Documentation

(d) Software testing

Reason: Quality Assurance is a procedure to ensure the quality of a product or service, and its process includes preparing criteria and comparing results against them; verification is checking against the defined criteria and standards, which is the assurance activity in that process.

3. Digital marketing is often referred to as _____?

(a) billboard

✓ (b) online marketing

(c) TV

(d) radio

Reason: Digital Marketing is a tool used to promote brands and to connect with potential customers online, so it is often referred to as online marketing.

4. Social Media Marketing is a type of _____?

(a) Digital Marketing

(b) Traditional Marketing

(c) Industrial Marketing

(d) Telemarketing

Reason: Social media marketing uses digital platforms such as Facebook, YouTube and Twitter to promote products or services, so it is a type of digital marketing.

5. Which of the following are the elements of marketing?

(a) Product, Price, Promotion and People

(b) Product, Price, Promotion and Place

(c) Product, Plan, Promotion and Place

(d) Product, Plan, Promotion and People

Reason: There are four elements of marketing: product, price, promotion and place.

Short Response Questions

Question: 1. Describe the basic components of business plan.

Answer: A business plan describes the company's goal, strategy and implementation for the growth of an organization. Its five components are: (1) Customer needs (market need) — what the customers actually need; (2) Product design — designing the product or service to meet that need; (3) Costing — working out the cost of making the product and the price at which it will be sold; (4) Operations — how the business will run day to day

to produce and deliver the product; and (5) Marketing — how the product will be promoted and sold to customers.

Question: 2. Describe the four elements of marketing.

Answer: Marketing is the action or business of promoting and selling products or services, including market research and advertising. Its four elements are: (1) Product — the goods and services the business offers to the end user, which must be worth purchasing against competitors' products; (2) Price — an important element decided keeping in mind the competitive price of the product; (3) Promotion — the packaging and position or place of the product in the marketplace where customers can be attracted, using marketing tools to promote sales; and (4) Place — the location of the outlet and the mode of selling, whether by e-commerce, telephone or retail outlets.

Question: 3. What is the difference between payment and transaction?

Answer: A transaction is an agreement between a buyer and a seller to exchange goods, services or financial assets in return for money; it is the act of conducting or carrying out business. Payment is the transfer of money against the purchase of goods or services; it is the act of paying within that business. In short, the transaction is the whole deal between buyer and seller, while the payment is the money part of it, which can be made by credit cards, debit cards, bank transfer or Easypaisa.

Question: 4. Why a quality assurance is important to assess the quality of a product or service?

Answer: Quality is all about meeting the needs and expectations of customers with respect to functionality, design, reliability, durability and price of the product, and assurance is a positive declaration on a product

or service which gives confidence to the user. Quality Assurance is important because it is the procedure that ensures the quality of products or services provided to customers: it focuses on improving the development process and making it efficient and effective as per the quality standards defined for products, through steps such as taking measures, identifying the problem area, identifying indicators, preparing criteria, comparing results and analysing assessment. It helps a company design and create a product which meets the expectations of customers and produces a high quality product that builds up the trust and loyalty of the customers.

Question: 5. Write down three advantages and three disadvantages of digital marketing.

Answer: Advantages of digital marketing: (1) Digital marketing has removed all sorts of geographical barriers, so products can be promoted worldwide. (2) Reduction in costs through automation and use of electronic media, with extremely low risk. (3) Faster response to both marketers and the end users, increased exposure of products or services, and winning customer loyalty. Disadvantages of digital marketing: (1) It depends on technology, so it fails where technology or connectivity is missing. (2) Security and privacy issues for users and businesses. (3) Maintenance cost is more due to a constantly evolving environment, along with higher transparency of pricing, increased price competition and worldwide competition through globalization.

Activity

Question: 1. Pakistan Airline (PIA) plans to offer incentives to their valuable customers. Chalk out a marketing strategy to promote PIA offer using digital platforms or social media marketing.

Answer: Marketing strategy for PIA's customer incentives, using the book's digital marketing tools: (1) Product and value proposition: present the incentive as an innovation that makes flying PIA more attractive, for example bonus miles, discounted tickets for frequent flyers, extra baggage allowance and priority check-in for valuable customers. (2) Digital platforms: promote the offer worldwide through digital platforms such as Facebook, YouTube and Twitter, plus PIA's website and mobile app, since a digital platform is an electronic tool to promote products or services worldwide. (3) Promotion: create awareness to increase sales with short videos of the offer, "Buy 1 Get 1 free"-style deals on selected routes, and social media marketing posts that draw the attention of end users and encourage immediate booking. (4) Place and payment: let customers book and pay online through the website and app using credit cards, debit cards, bank transfer or Easypaisa, so the whole transaction from offer to ticket is digital. (5) Search Engine Optimization: improve PIA's website visibility so people searching for flights and offers in Google, Bing and other search engines find the incentive offer first.